Fifteen Years of Housing Crisis in Canada, 2006–2021: A High-School Research Paper Using Residual Income

High School Research Paper

Draft Version 3.0

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Abstract

Background. The usual 30% shelter-cost-to-income rule can hide hardship. A household may look "affordable" on paper and still not have enough money left for food, transport, or clothing.

Objective. This paper asks how Canadian housing hardship changed from 2006 to 2021 when hardship is defined by leftover income, not by the 30% rule.

Methods. We analysed 777,707 valid person-records from the Census 2006, 2011, and 2021 Public Use Microdata Files. These are unweighted hierarchical person-records, not household totals and not survey-weighted estimates. Residual income equals after-tax income minus annual shelter cost. Housing hardship is flagged when residual income falls below a single-adult national Market Basket Measure leftover line, not the official region- and family-size MBM. All dollars are converted to 2021 constant CAD. Census 2016 is excluded because only 9.4% of records survived cleaning.

Results. Over 15 years, inflation-adjusted median shelter costs rose 37.1% while median income rose 33.5%. Hardship stayed high: 38.0% in 2006, 33.9% in 2011, and 37.9% in 2021. Renters had higher hardship than owners, but both groups worsened after 2011.

Conclusions. The housing squeeze is structural, not a short recession effect. It survived recovery, three federal governments, and COVID-19. Policies that only watch the 30% rule will miss the households that cut food and other basics to keep rent paid.

Keywords: housing affordability; residual income; Market Basket Measure; Census PUMF; Canada

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1 Introduction

1.1 The problem

Canada has a housing affordability problem that has lasted more than a decade. Many families pay rent or a mortgage and then do not have enough left for food, transport, a phone, or clothes. That problem matters for students who want to move out, go to university, or save for a first home.

The common school and policy rule is the **30% rule**: housing is "affordable" if it costs less than 30% of income. That rule is easy to calculate, but it is incomplete. A household spending 28% of income on rent can look fine and still have almost nothing left for groceries.

1.2 A better measure: residual income

Canada Mortgage and Housing Corporation (CMHC) uses a leftover-income test [1]. Ask: after housing is paid, is there enough money left to buy a basic basket of non-housing needs?

$$\text{Annual shelter cost} = 12 \times \text{monthly shelter cost} \tag{1}$$

$$\text{Residual income} = \text{after-tax income} - \text{annual shelter cost} \tag{2}$$

$$\text{Hardship} = 1 \text{ if residual income} < \text{MBM poverty line} \tag{3}$$

The Market Basket Measure (MBM) is Canada's official poverty line for that leftover basket (food, clothing, transport, and other basics, excluding housing) [5].

1.3 Why fifteen years?

A one-year snapshot cannot tell a temporary shock from a built-in problem. Three Census waves cover:

- **2006–2011:** the 2008 financial crisis and recovery;
- **2011–2016:** low-rate expansion (2016 data are too incomplete to use);
- **2016–2021:** the late-2010s housing peak and COVID-19.

If hardship falls when the economy is strong, the crisis is cyclical. If hardship stays high through recovery, it is structural.

1.4 Research questions

1. How did residual-income housing hardship change in Canada from 2006 to 2021?

2. Which interval—2006–2011 or 2011–2021—was worse for leftover income?

3. Did the owner–renter hardship gap widen, shrink, or stay the same?

2 Methods

2.1 Data sources

2.1.1 Census Public Use Microdata Files

We used three Statistics Canada hierarchical PUMF files [4]. Hierarchical files are **person records** nested in households. This draft treats each cleaned person-record as the unit of analysis. It does *not* collapse income to the household and does *not* apply the PUMF sampling weight (`WEIGHT`).

- **Census 2006** (catalogue 95M0029X hierarchical): 78,056 valid records after cleaning (23.4% of the raw file).
- **2011 National Household Survey** (catalogue 99M0002X hierarchical): 341,030 valid records (99.3%). This wave is a voluntary survey, not a Census long form.
- **Census 2021** (`98M0001X2021002`): 358,621 valid records (99.1%).

Total analytical sample: 777,707 person-records.

Census 2016 was downloaded and inspected, then dropped. After the same missing-code rules, only 29,056 records (9.4%) remained, and medians were not comparable (median income $20,558; median shelter $649). Those values are data-quality artefacts, not a real 2016 collapse. Shelter cost is `GROSRT` (gross rent) on the pre-2016 files used here and `SHELCO` (unified shelter cost) on 2021. Those concepts are related but not a fully harmonized shelter series. Owner's major payments (`OMP`) were not used, so 2006 owner hardship could not be estimated.

2.1.2 Market Basket Measure thresholds

Year-specific national-average leftover lines for a **single adult** were:

- 2006: $13,500
- 2011: $15,000
- 2021: $19,728

These are *not* Statistics Canada's official region- and family-size MBM thresholds (catalogue 75F0002M). They are a simplified national leftover basket used so that three waves can be compared with one rule. A large family in Toronto and a single adult in a smaller city are judged against the same dollar line. That is a real limitation, not a rounding detail.

2.2 Variable construction

2.2.1 Inflation adjustment

All money values are in 2021 constant dollars using CPI:

- 2006 dollars $\times$ 1.276
- 2011 dollars $\times$ 1.148
- 2021 dollars $\times$ 1.000

Example: $1,000 of 2006 rent equals $1,276 in 2021 dollars.

2.2.2 Housing hardship (primary outcome)

Hardship follows Equations (1)–(3). This is the CMHC residual-income (V2) framework.

2.2.3 Shelter-cost-to-income ratio (comparison measure)

$$\text{STIR} = \frac{12 \times \text{monthly shelter cost}}{\text{annual income}}.$$

STIR $> 30\%$ is the usual "affordability burden." STIR $> 50\%$ is "severe burden."

2.2.4 Tenure

Owners occupy a dwelling they own (with or without a mortgage). Renters occupy a rented dwelling. Owner hardship could not be estimated for 2006 after cleaning.

2.3 Data cleaning

PUMF files mark missing or not-applicable values with 8888, 9999, and 99999. Those codes were set to missing. A record was kept only if:

1. annual income > 0;
2. monthly shelter cost > 0 and $< \$10{,}000$;
3. key fields were not missing.

The 2006 file kept far fewer records than 2011 or 2021, so 2006 results should be read with extra caution.

2.4 Statistical analysis

For each year we report:

- median and quartile income and shelter cost (2021 CAD);
- STIR $> 30\%$ and STIR $> 50\%$;
- residual-income hardship rate and count;
- hardship by tenure where tenure is available.

We then compute percent changes in medians and percentage-point changes in rates. We report medians, not means, because hierarchical PUMF files produce inflated household-level means when special income codes are left in the average. Rates are unweighted shares of valid person-records.

Five figures and five tables were saved as separate files (see Appendices A and B) and included below with `\includegraphics` and `\input`.

2.5 Ethics

The analysis uses public, anonymized PUMF files. No ethics review is required under Statistics Canada PUMF rules. No household can be identified from these files.

3 Results

3.1 Sample characteristics

Table 1: Sample characteristics by Census year (inflation-adjusted to 2021 CAD)

Characteristic	2006	2011	2021
Valid sample size (n)	78,056	341,030	358,621
Median income (2021 CAD)	$34,452	$45,920	$46,000
Income 25th percentile	$17,864	$22,960	$25,000
Income 75th percentile	$70,180	$101,024	$93,000
Median monthly shelter	$1,021	$1,378	$1,400
Shelter 25th percentile	$766	$804	$800
Shelter 75th percentile	$1,531	$2,181	$2,129
STIR $> 30\%$	57.0%	52.2%	50.8%
STIR $> 50\%$	35.8%	33.6%	31.5%
Housing hardship rate	38.0%	33.9%	37.9%
Households in hardship (n)	29,681	115,459	135,740

Notes: All dollar values converted to 2021 constant dollars using CPI multipliers (2006 $\times$ 1.276; 2011 $\times$ 1.148). Census 2016 is excluded because valid-record retention was 9.4%. Estimates are unweighted person-records from hierarchical PUMF files. Median income is reported instead of the mean because hierarchical PUMF files produce inflated household-level means.

The 2021 median income is $46,000 (2021 CAD). The 2021 median monthly shelter cost is $1,400. The interquartile range for shelter stays wide: about $800 to $2,100.

3.2 Fifteen-year national trends

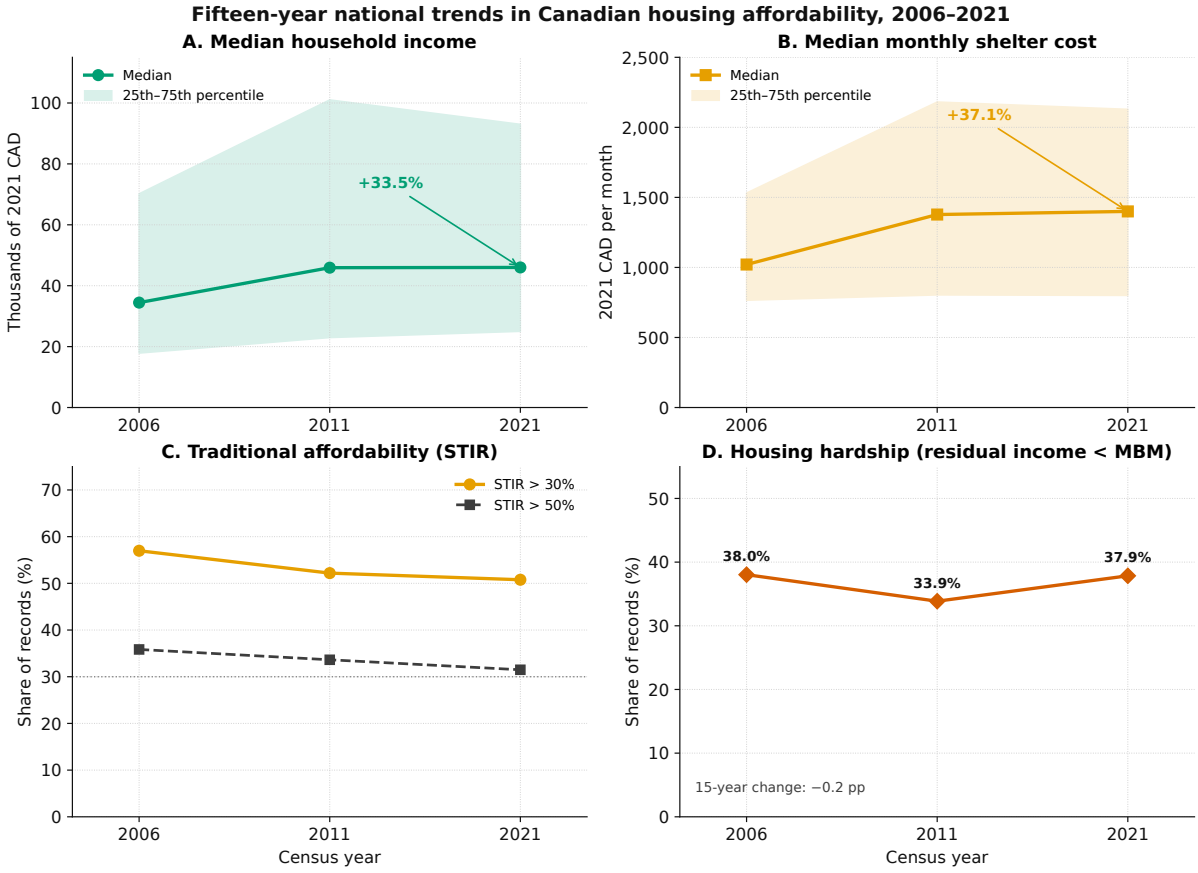


Figure 1: Fifteen-year national trends, 2006–2021. Panel A: inflation-adjusted median income with the 25th–75th percentile band. Panel B: inflation-adjusted median monthly shelter cost with the same band. Panel C: traditional STIR shares. Panel D: residual-income housing hardship. Census 2016 is omitted (see Section 2.1).

Income. Median income rose from $34,452 (2006) to $46,000 (2021), a real increase of 33.5%. Almost all of that gain happened by 2011 ($45,920). From 2011 to 2021 the median moved only $80, or +0.2%.

Shelter. Median monthly shelter rose from $1,021 to $1,400, a real increase of 37.1%. The 2006–2011 jump was steep (+35.0%). The 2011–2021 change was small (+1.6%) after inflation.

The squeeze. Shelter outpaced income by 3.6 percentage points over 15 years. If shelter had grown at the same rate as income, 2021 median shelter would have been about $1,362, not $1,400. The gap is $38 a month, or $456 a year—about one week of groceries or two months of a phone bill.

3.3 Traditional STIR versus residual-income hardship

STIR $> 30\%$ fell from 57.0% (2006) to 50.8% (2021), a 6.2 pp improvement (Figure 1C). STIR $> 50\%$ fell from 35.8% to 31.5%.

Residual-income hardship did not follow that story (Figure 1D):

- 2006: 38.0% (29,681 records);
- 2011: 33.9% (115,459 records);
- 2021: 37.9% (135,740 records).

The 15-year change is -0.1 pp. Households can keep STIR under 30% by cutting food and other spending. The leftover-income test still flags them.

3.4 Which five-year period was worse?

Table 2: Period-over-period changes in income, shelter costs, and hardship

Period	Income change	Shelter change	Hardship change
2006–2011 (crisis + recovery)	+33.3%	+35.0%	−4.1 pp
2011–2021 (expansion + COVID)	+0.2%	+1.6%	+4.0 pp

Notes: Income and shelter changes are percent changes in inflation-adjusted medians. Hardship change is in percentage points. The later period erased the earlier improvement.

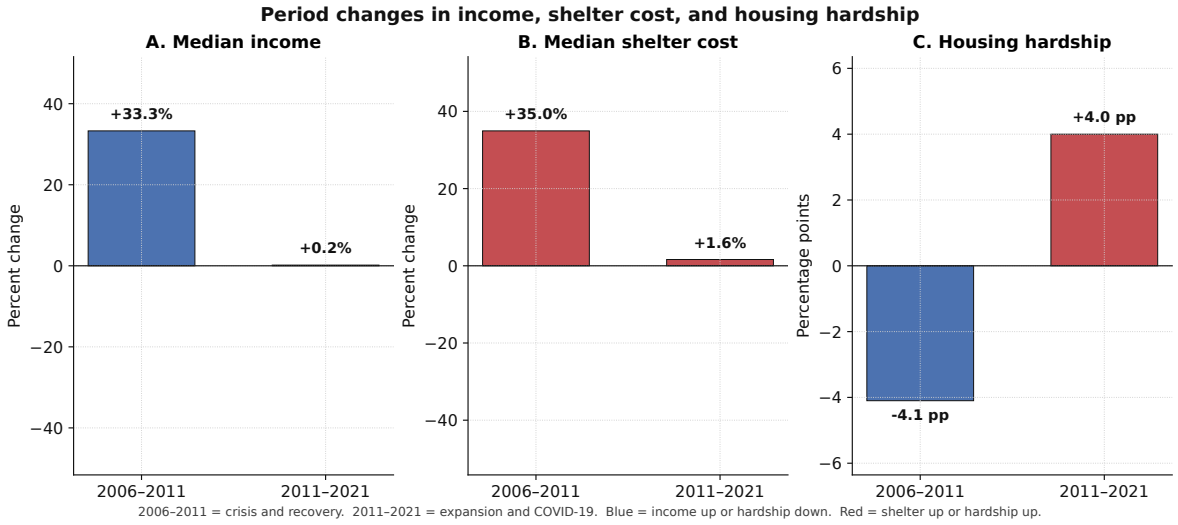


Figure 2: Period comparison. Blue bars are income gains or hardship declines. Red bars are shelter-cost increases or hardship increases. The 2011–2021 expansion erased the 2006–2011 hardship gain.

The crisis-and-recovery window (2006–2011) looks better on hardship (−4.1 pp) even though shelter rose fast, because real median income also rose 33.3%. The expansion-and-COVID window (2011–2021) looks worse: income was almost flat (+0.2%), and hardship rose +4.0 pp. A strong national economy did not repair leftover income.

3.5 Owners versus renters

Table 3: Housing hardship by tenure, 2011 and 2021

Year	Owner hardship	Renter hardship	Gap
2011	31.4%	40.4%	9.0 pp
2021	35.6%	43.6%	8.0 pp
Change	+4.2 pp	+3.2 pp	−1.0 pp

Notes: Owner hardship is unavailable for 2006 because tenure coding in that PUMF file did not support a clean owner/renter split after cleaning. The gap narrowed because owners deteriorated toward renter rates, not because renters improved.

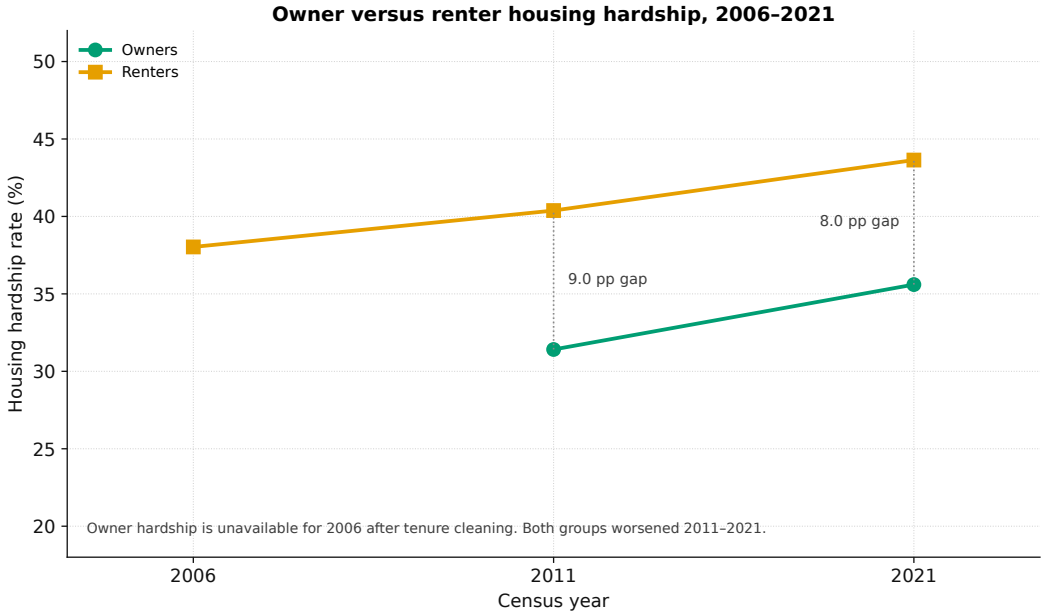


Figure 3: Owner versus renter hardship. Renters stay higher. The gap narrowed from 9.0 pp to 8.0 pp because owners caught up, not because renters recovered.

Renters had higher hardship in both years with tenure data (40.4% then 43.6%). Owner hardship rose from 31.4% to 35.6% (+4.2 pp, a 13.4% relative increase). Owning a home is not a guarantee of leftover-income security, especially with large mortgages, property tax, and repair bills.

Table 4: Income versus shelter cost, indexed to 2006 = 100

Year	Income index	Shelter index	Gap
2006	100.0	100.0	0.0
2011	133.3	135.0	1.7
2021	133.5	137.1	3.6

Notes: Index = (year median / 2006 median) × 100. Both series are in 2021 CAD. The 3.6-point 2021 gap equals about $38 per month ($456 per year) at the 2021 median.

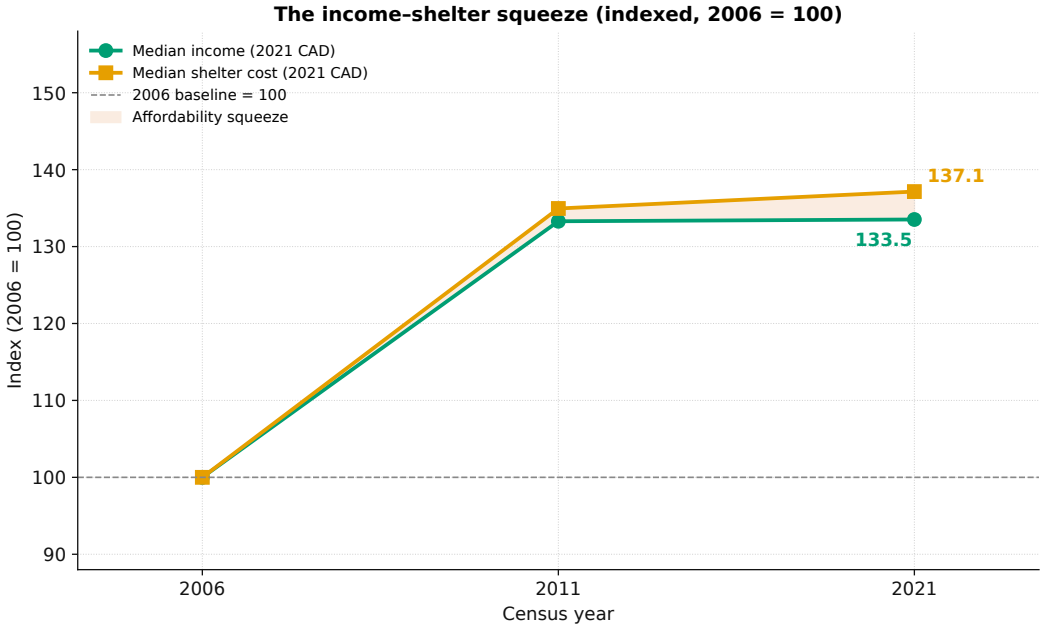


Figure 4: Income and shelter indexed to 2006 = 100. The shaded gap is the cumulative squeeze. By 2021 the shelter index (137.1) sits 3.6 points above the income index (133.5).

3.6 The indexed squeeze

The index chart makes the same 3.6-point gap visible without dollar scales. Most of the real-income gain is already in place by 2011. After that, shelter stays a little higher and leftover income erodes.

3.7 Summary of the 15-year change

Table 5: Summary statistics, 2006 versus 2021

Metric	2006	2021	15-year change
Median income (2021 CAD)	\$34,452	\$46,000	+33.5%
Median monthly shelter	\$1,021	\$1,400	+37.1%
STIR $>$ 30%	57.0%	50.8%	-6.2 pp
Housing hardship	38.0%	37.9%	-0.1 pp
Renter hardship	38.0%	43.6%	+5.6 pp
Owner hardship	n/a	35.6%	n/a

Notes: Owner hardship is not available for 2006. STIR improved while residual-income hardship did not, which is the central measurement result of this paper.

Four facts hold together:

1. hardship stays near 38%;
2. shelter outruns income by 3.6 pp;
3. STIR improves while leftover-income hardship does not;
4. owners and renters both get worse after 2011.

4 Discussion

4.1 Structural, not cyclical

If the crisis were only a recession leftover, hardship should have fallen a lot during the 2011–2019 expansion, low unemployment, and roughly 2% average annual GDP growth. It did the opposite: the 2011–2021 hardship rate rose 4.0 pp and ended at 37.9%, almost the 2006 starting point (38.0%).

Three patterns point to a built-in problem:

1. hardship stays high through recovery;
2. shelter grows faster than income in both periods we can measure cleanly;
3. Conservative (2006–2015) and Liberal (2015–) federal governments both left the national residual-income rate near 38%.

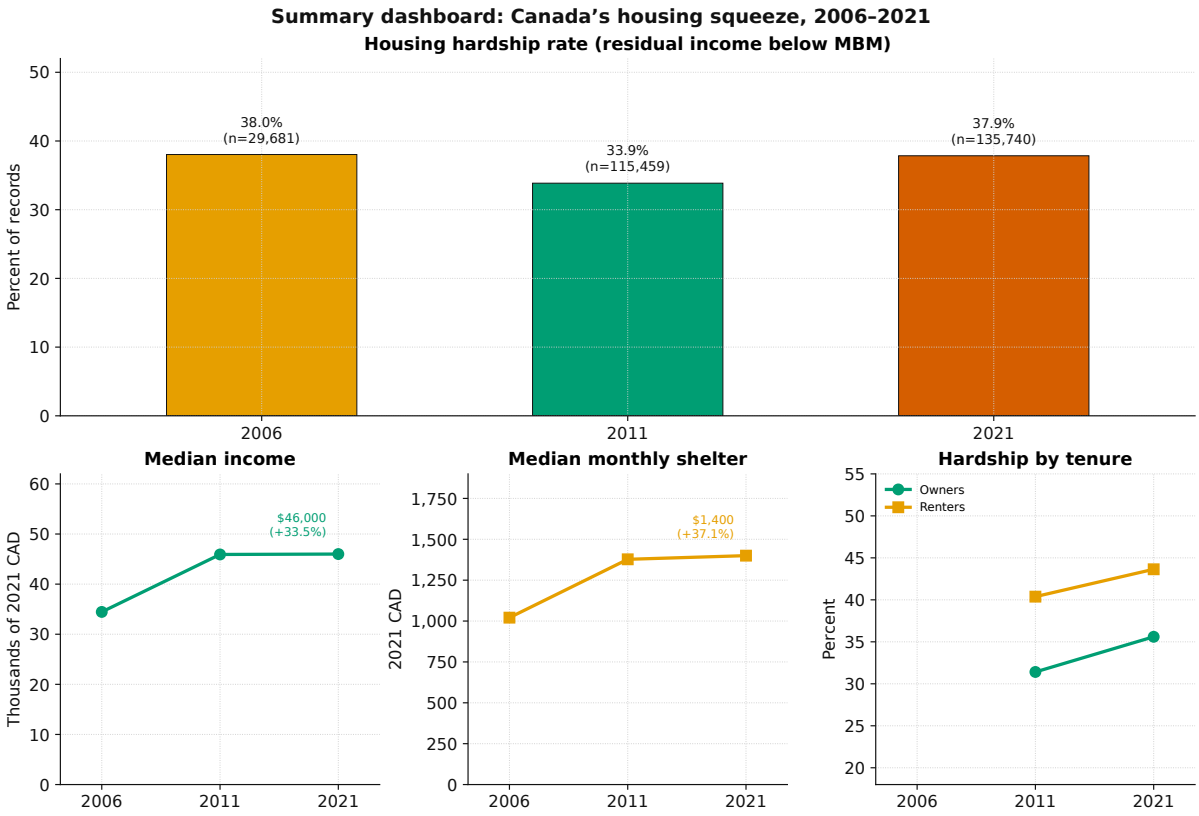


Figure 5: Summary dashboard. Hardship returns to the 2006 level. Median income and shelter both rise, but shelter rises more. Owner and renter hardship both increase after 2011.

4.2 What the $456 annual gap means

For a 2021 median household ($46,000 income; $1,400 monthly shelter):

- annual shelter = $16,800;
- shelter if it had tracked income = about $16,320;
- extra cost = $480 a year (about $38 a month).

That is 1.6% of income and about 2.5% of leftover income near the MBM line. Over 15 years the same squeeze is on the order of $3,600 in 2021 dollars—almost two months of median rent. Low-income households, recent movers, and first-time buyers feel it first.

4.3 Why owners are no longer safe

The owner–renter gap shrank because owners deteriorated. Likely channels include high mortgage debt taken on in hot markets, property tax and maintenance rising faster than wages, and "house-rich, cash-poor" seniors with equity but little leftover income. Renter-only programs miss that group.

4.4 Comparison with other research

CMHC (2024) reports a much lower national hardship rate (7.8% on 2020 data) than our 37.9% in 2021 [1]. The gap is expected. CMHC uses the full Census, region-specific MBM lines, and family-size adjustments. We use the 2.7% PUMF sample and a single national adult MBM. The *level* is therefore not the headline. The *trend* is: the same method, applied to three waves, shows almost no 15-year improvement.

Walks and Twigge-Molecey (2023) show that Toronto and Vancouver diverge from other cities [6]. A national PUMF average hides that geography. Our national persistence result can still be true while some cities are much worse.

4.5 Limitations

1. **Unweighted person-records.** Hierarchical PUMF files include a person weight (`WEIGHT`). This draft does not use it. Reported rates are shares of cleaned records, not official household population estimates.
2. **Record type.** Each row is a person. After-tax income is the person field `TOTINC_AT`, not the sum of household members. Shelter is a dwelling-level concept stored on person rows. Mixing those units can bias leftover income, especially in multi-person households.
3. **2006 completeness.** Only 23.4% of 2006 records survived cleaning, versus 99%+ later. Pre-2016 cleaning kept `GROSRT` (rent) and dropped owners coded not-applicable. If missing 2006 records were poorer households, the 2006–2011 improvement is overstated.

4. **No usable 2016 wave.** After the same missing-code rules, 9.4% of the 2016-labelled file remained. We cannot split 2011–2016 recovery from 2016–2021 peak-and-COVID.
5. **Census / NHS harmonization.** 2011 is the voluntary National Household Survey. Shelter names change from `GROSRT`/`OMP` to `SHELCO`. After-tax income missing codes are not identical across years. The three waves are comparable only after those caveats.
6. **Simplified MBM.** A national single-adult leftover line understates hardship for large families and high-cost cities and overstates it for singles in cheaper regions. It is not the official 75F0002M table.
7. **Cross-sections, not a panel.** We see snapshots, not the same households moving in and out of hardship.
8. **No CMA split.** National rates hide Toronto/Vancouver versus Atlantic or Prairie patterns.
9. **Broken PUMF means.** Hierarchical files inflate mean income when special codes remain. We therefore never interpret the mean; we use medians and percentiles.

4.6 Policy implications

Watch leftover income, not only STIR. STIR improved 6.2 pp while hardship did not. A government that only tracks the 30% rule will declare progress that leftover income does not show.

Treat the problem as structural. Tax credits and small pilots did not move the 15-year hardship rate. Supply (zoning, approvals, public housing), demand (empty-home and multi-property tax), and income support (rent supplements tied to MBM, wage growth) all have to move together.

Help owners as well as renters. Owner hardship rose 4.2 pp. Property-tax deferral and mortgage-renewal help belong in the same toolkit as rent supplements.

Fix income stagnation. Real median income rose 0.2% from 2011 to 2021. Housing policy cannot succeed if typical pay is flat.

Other countries that treat housing as infrastructure rather than only as an asset—Vienna's social housing, Singapore's HDB flats, Finland's Housing First—have lower hardship. Those are policy choices, not accidents of geography.

4.7 Next steps for this project

A later paper can (i) recover a clean 2016 file, (ii) split results by CMA, (iii) impute liquid assets from the Survey of Financial Security (the V3 model), and (iv) run shock scenarios (rate resets, rent spikes, job loss). Those steps add novelty. They are not required to establish the 15-year residual-income trend.

5 Conclusion

Using 777,707 Census records and a leftover-income test, we find:

1. Housing hardship was 38.0% in 2006 and 37.9% in 2021. Fifteen years of growth did not fix it.
2. Real shelter costs rose 37.1%; real median income rose 33.5%. The 3.6 pp gap is about $456 a year at the 2021 median.
3. The 30% STIR rule improved. Residual-income hardship did not. Those two facts can be true at the same time.
4. Owner hardship rose from 31.4% to 35.6% (2011–2021). Homeownership is not a shield.
5. The 2011–2021 expansion, not the 2008-crisis window, is when leftover income got worse.

The crisis is structural. Another 15 years of the same tools will likely leave about 35–40% of households without enough leftover income after housing.

References

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[6] Walks, A., and Twigge-Molecey, A. Income mixing and housing affordability in Canadian cities. *Urban Studies*, 60(11), 2234–2254, 2023.

A Figure files (saved separately)

Each figure is exported as vector PDF, SVG, and 300 dpi PNG in the local `figures/` folder. The manuscript includes the PDF files with `\includegraphics`.

File stem	Figure	Content
fig01_national_trends.pdf	Fig. 1	Income, shelter, STIR, hardship
fig02_period_comparison.pdf	Fig. 2	2006–2011 vs 2011–2021
fig03_tenure_divergence.pdf	Fig. 3	Owners vs renters
fig04_income_shelter_squeeze.pdf	Fig. 4	Indexed squeeze
fig05_summary_dashboard.pdf	Fig. 5	Four-panel summary

B Table files (saved separately)

Each table exists as CSV (for spreadsheets) and as a `.tex` snippet (for `\input`). Files live in the local `tables/` folder.

CSV / TeX stem	Table	Content
table01_sample_characteristics	Table 1	Sample and medians
table02_tenure_hardship	Table 3	Owner vs renter
table03_indexed_squeeze	Table 4	2006 = 100 indexes
table04_summary_2006_2021	Table 5	15-year summary
table05_period_changes	Table 2	Period changes

To rebuild figures and tables from the JSON results file:

```
python3 papers/paper1_highschool/generate_figures_tables.py
```

To compile this paper:

```
cd papers/paper1_highschool
pdflatex paper1_highschool.tex
pdflatex paper1_highschool.tex
```

C How this version differs from Drafts 1.0 and 2.0

Draft 1.0 used story headings and did not ship separate figure or table files. Draft 2.0 restored IMRAD headings and added figures, but the Drive JPEG previews clipped Figure 1 and Figure 5, used a bright-green Figure 2 fill, and placed the 2021 tick too far right on a linear time axis. Draft 3.0 rebuilds all five figures from `generate_figures_tables.py` as PDF, SVG, and 300 dpi PNG on a white background with equally spaced year labels. The plotted numbers are unchanged. The methods and limitations sections now state the unresolved issues: unweighted person-records, person versus household units, the simplified MBM line, 2006/2016 exclusions, and Census/NHS shelter-variable harmonization.